

Roll No.

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Paper Code: TCS 302

Mid Semester Examination 2023
B.Tech (CSE-Core/CSE-Specialization) III Semester
Data Structure with 'C' language.

Time: 1:30 Hrs

MM: 50

INSTRUCTIONS TO STUDENTS

Note :

- (i) This question paper contains five questions with alternative choice.
- (ii) All questions are compulsory.
- (iii) Each question carries **two parts a or b**. Attempt either parts **a** or **b** of each question.
- (iv) Total marks assigned to each question are ten.

Q1.

(10*1,CO1,CO2)

A. Assume that you have a single linked list, first node of the linked list is pointed by a pointer Ptr. Write a C function to print the linked list in reverse order.

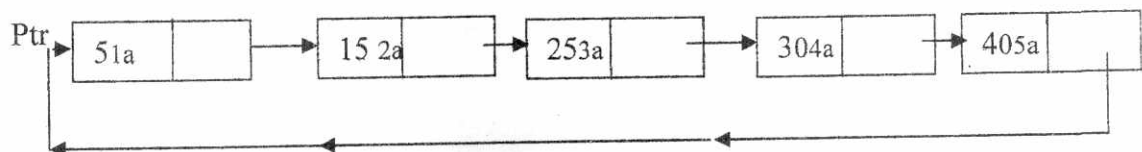
OR

B. Convert the following infix expression into postfix expression using stack (show all steps). $X*Y-(Z+W)/T-Q\%M$

Q2.

(10*1,CO1,CO3)

A. Consider a Circular linked list with a pointer, Ptr. Write a C function to insert a new node before the node pointed by pointer Ptr, in the linked list.



OR

B. Assume that we have a singly linked list, first node of the linked list is pointed by a pointer PTR. Write a C function to split that linked list in two linked lists as per the choice given by the user(Node 's information).

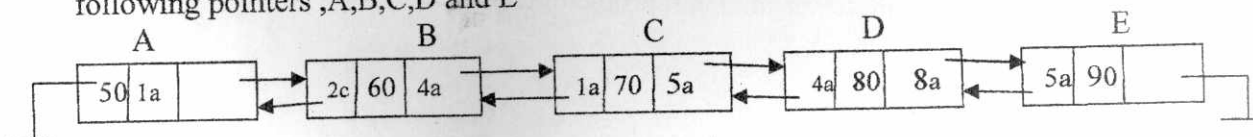
Q3.

(10*1, CO1,CO2)

A. What do you mean by a dynamic array? Write a 'C' function to create a dynamic array to store N elements and then check whether the sequence of elements is in A.P or not.

OR

B. Assume that you have a double linked list, with five nodes, nodes are pointed by the following pointers ,A,B,C,D and E



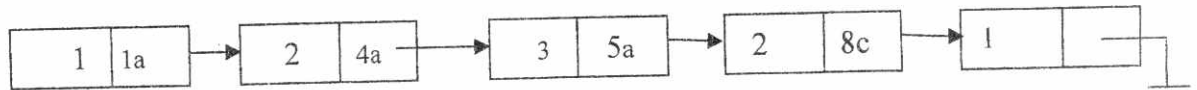
Write steps to insert a node between the nodes pointed by pointers C and D.

Q3.

(10*1,CO1,CO2)

- A. Assume that we have a single linked list with a pointer P at first node. Write a c function to check whether the linked list is palindrome or not.

Example :



The given linked list is a palindrome.

OR

- B. Assuming that you have a single linked list, first node of the linkedlist is pointed by a pointer ptr. Write a C function to delete a node from the linked list as per the choice given by the user.

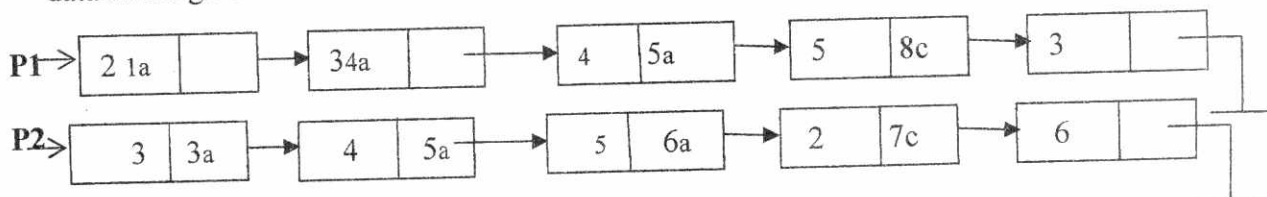
Q4.

(10*1, CO1,CO2)

- A. Assuming that you have a single linked list, first node of the linked list is pointed by a pointer Ptr. Write a C function to count nodes having odd information in the linked list.

OR

- B. Assume that you have two single linked lists , pointers P1 and P2 are pointing to the first node of the linked lists respectively. Write a C function to print the multiplication of the data in the given linked lists.



Output: 6 , 12,20 ,10 18

Q5.

(10*1, CO2,CO3)

- A. Assume that we have a single linked list. First node is pointed by pointer P. Write a C function to find and print the middle node in the linked list.

OR

- B. Explain tower of Hanoi problem. Draw recursive tree for 3 discs movement.

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TCS-307

B. TECH. (CSE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
OBJECT ORIENTED PROGRAMMING WITH C++

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.
(iii) For programming questions provide typical output.

1. (a) Discuss OOP features with examples. How are these implemented in terms of software design in C++ ? (CO1)

OR

- (b) Explain the need of namespaces in C++ with the help of a real-world example. Also implement the real-world example that you have assumed with the help of proper C++ code. (CO1)
2. (a) Construct a C++ program to input a word from the console and replace the consecutive duplicates by its single occurrence. (CO1)

TYPICAL INPUT : PROOOOGGRAMMMMIING

TYPICAL OUTPUT : PROGRAMING

P. T. O.

OR

- (b) Discuss inline function in C++ along with its advantages and disadvantage. Implement a program in C++ to find the greatest among two number using inline function. (CO1)
3. (a) Implement a C++ program by creating a class TestDemo with the following members :
- Data Members :
- String str1
- String str2
- Member Functions :**
- Constructor to initialize value of str1 and str2
- Void show () : to print value of str1 and str2
- Void Swap () : to swap value of str1 and str2
- In main function create object of class TestDemo and perform the above task. (CO2)

OR

- (b) Implement a class Employee with the help of a C++ program to read and display the necessary information of 'n' employees. The necessary data members of this class are as follows : (CO2)
- Data members :
- emp_name, emp_code, emp_dept and emp_salary (all private)
- Member function :
- Public : display info ()
- Read the details of 'n' employees by using the concept of array of object using constructor only.

(3)

4. (a) Differentiate static and non-static member function in C++. Explain with the help of C++ program the use of this pointer and why it is required. (CO2)

OR

- (b) Discuss why do we require a friend function, Construct a C++ program to swap the value of data members present in two different classes with the help of friend function. (CO2)
5. (a) Illustrate the need of operator overloading in C++ ? Implement a C++ to overload the binary operator the type $ob2=10+$, where $ob1$ is an object of class Complex that contain two arguments (real, imaginary), here 10 is interpreted as a real part only ($10+0*i$) i.e., no imaginary part. The sequence of the statement $ob2=10+ob1$ should not be changed.

(CO2)

OR

- (b) Implement a C++ program to overload Post decrement operator (- -) using friend function only. (CO2)

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TCS-308

B. TECH. (CSE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
LOGIC DESIGN AND COMPUTER ORGANIZATION

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Simplify the following expression using K-Map method : (CO1)

$$F(A, B, C, D, E) = \sum m (5, 7, 8, 9, 10, 11, 13, 15, 21, 23, 25, 26, 27, 29, 31)$$

OR

- (b) Simplify the following expression using Quine McCluskey method :

(CO1)

$$F(A, B, C, D) = \sum m (1, 3, 7, 11, 15) + \sum d (0, 2, 5)$$

2. (a) Draw and explain priority encoder. (CO1)

OR

- (b) Implement the following expression using 4 : 1 MUX when C and D are selected lines : (CO1)

$$F(A, B, C, D) = \sum m (1, 3, 5, 7, 10, 11, 14, 15)$$

P. T. O.

(2)

3. (a) Implement 3 bit binary to gray code converter using suitable decoder.

(CO1)

OR

- (b) Implement 1 bit magnitude comparator using 4 : 1 MUX. (CO1)

4. (a) Draw and explain 4 bit by 3 bit binary multiplier. (CO1/CO2)

OR

- (b) What are the differences between Latch and flip-flop and convert D flip-flop in to T flip-flop. (CO1/CO2)

5. (a) Explain and draw 4 bit parallel in serial out shift register. (CO3)

OR

- (b) Design Mod-16 up ripple counter. (CO3)

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TMA-316

B. TECH. (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

DISCRETE STRUCTURES AND COMBINATORICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Let $A = \{1, 2, 3\}$ and let R be a relation defined on A as : $R = \{(1, 1), (2, 2), (1, 2), (2, 3), (3, 3)\}$. Is R :

(i) Reflexive

(ii) Symmetric

(iii) Transitive

(iv) An equivalence relation ?

Draw its diagram and also find the relation matrix M_R . Compute

$R^2 = R \circ R$. (CO1)

OR

(b) Let $A = \{1, 2, 3, \dots, 9\}$ and R be the relation in $A \times A$ defined by $(a, b) R (c, d)$ if $a + d = b + c$ for $(a, b), (c, d)$ in $A \times A$. Prove that R is an equivalence relation. (CO1)

P. T. O.

2. (a) Show that the function $f : \mathbb{R} \rightarrow (1, \infty)$ and $g : (1, \infty) \rightarrow \mathbb{R}$, defined by $f(x) = 5^{3x} + 1$, $g(x) = (1/3)\log_5(x - 1)$ are inverses. (CO1)

OR

- (b) The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = 2x^3 - 1$. Show that f is invertible and find the inverse of f . (CO1)
3. (a) Prove by the principal of mathematical induction that $11^{n+2} + 12^{2n+1}$ is divisible by 133 for each positive integer n . (CO3)

OR

- (b) Draw the Hasse diagram of D_{36} . Let the sets $A = \{9, 18\}$ and $B = \{4, 6, 9\}$. Find, if they exist : (CO1)
- (i) Upper bounds of A and B ;
 - (ii) Lower bounds of A and B ;
 - (iii) Supremum (lub) of A and B ;
 - (iv) Infimum (glb) of A and B ;
4. (a) (i) If $5x + 1 = 11$, then $x = 2$. Find its inverse, converse and contrapositive.
- (ii) Verify that $p \rightarrow q \equiv \sim p \vee q$. (CO3)

OR

- (b) Find which of these is a tautology or contradiction : (CO3)
- (i) $(p \vee q) \wedge (\sim p \wedge \sim q)$ and
 - (ii) $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$

(3)

5. (a) Show that $\sim p$ is a valid conclusion from the premises $p \rightarrow \sim q$,
 $\sim q \rightarrow r, \sim r$: (CO3)

(i) with truth table

(ii) without truth table

OR

- (b) Check the validity of the following argument :

“It is not sunny today and it is colder than yesterday. We will go swimming only if it is sunny. If we do not go swimming then we will take a boat trip. If we take a boat trip then we will be home by sunset. Therefore, we will be home by sunset.” (CO3)

(3)

5. (a) Show that $\sim p$ is a valid conclusion from the premises $p \rightarrow \sim q$,
 $\sim q \rightarrow r, \sim r$: (CO3)

(i) with truth table

(ii) without truth table

OR

- (b) Check the validity of the following argument :

“It is not sunny today and it is colder than yesterday. We will go swimming only if it is sunny. If we do not go swimming then we will take a boat trip. If we take a boat trip then we will be home by sunset. Therefore, we will be home by sunset.” (CO3)

Roll No.

TCS-331

B. TECH. (CSE) & B. TECH. (BIOTECHNOLOGY)

(THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

FUNDAMENTAL OF IoT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Discuss the historical evolution of internet of things and brief the key milestones. (CO1)

OR

- (b) A company utilizes the Internet of Things technology for building application and resources for potential smart home systems. Discuss the major challenges to be faced by the company and solutions for potential mitigation of the challenges. (CO1)

2. (a) What are the key differences between automation and internet of things ? Discuss it in a detailed comparison. (CO1)

P. T. O.

OR

(b) Discuss any *three* IoT terminologies : (CO1)

(i) Machine to Machine (M2M)

(ii) Gateway

(iii) Firmware

(iv) Telemetry

3. (a) Imagine you are an architect tasked with designing a smart city infrastructure that integrates various IoT devices for efficient traffic management. How would you plan the structural aspects of this IoT network to ensure seamless communication, data security, and scalability ? (CO2)

OR

(b) What do you mean by interoperability and scalability ? Discuss their significance in Internet of Things technology. (CO2)

4. (a) Discuss network topologies used in IoT systems along with their advantages and disadvantages. (CO2)

OR

(b) Bosch provides the products and services for Drive and Control Technology, Packaging Technology, Power tools for Professionals, Security Solutions, and Software Solutions. Bosch provides the IoT Suite for connecting and managing devices, sensors, and gateways. It provides secure access management. It also provides an open IoT platform for all domains. Classify the IoT sensors based on input, output and datatype in the perspective of Bosch. (CO2)

(3)

5. (a) Consider yourself an IoT engineer suggest the appropriate resources required for building a smart agriculture system with respect to different layers of IoT architecture. (CO2)

OR

- (b) Discuss the role of edge computing, fog computing, and cloud computing in IoT frameworks. (CO2)

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TCS-332

B. TECH. (CSE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

FUNDAMENTAL OF INFORMATION SECURITY AND BLOCKCHAIN

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the following approaches for effective cyber security : (CO1)
Confidentiality and Integrity.

OR

- (b) What is the zero trust model ? What are the different key steps of the zero trust model ? (CO1)

2. (a) Describe the following pillars of cyber security : (CO2)
Processes and Technology.

OR

- (b) Write down a shell script to copy a file from one location to another location. (CO2)

P. T. O.

(2)

3. (a) Write down a python program which creates a file that contains the entries like username and hashed password for 20 users. The username and password information should be taken from the user as input. (CO5)

OR

- (b) What is penetration testing ? Explain the different phases of penetration testing. (CO5)

4. (a) Demonstrate the security mechanisms which are available for the unauthorized accesses. (CO3)

OR

- (b) Explain the uses of the following Linux commands with their syntax: touch, echo, and chmod. (CO3)

5. (a) How would you compare the following categories of hackers ? (CO6)
Black hat hacker, White hat hacker and Gray hat hacker.

OR

- (b) Construct an access matrix for the following scenario :

- There are four users (i.e., U1, U2, U3 and U4) in the system.
- There are four objects/files (i.e., F1, F2, F3 and F4) in the system.
- User U1 cannot access F1, can write in F2, can read/write F3 and can execute F4.
- User U2 can read F1 and F2 and can read/write F3 and F4.
- User U3 can read/write F1 and F2 and can read/write/execute F3 and F4.
- User U4 can read/write/execute all files.

Roll No.

TCS-341

B. TECH. (CSE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
PYTHON PROGRAMMING FOR COMPUTING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any one sub-question among (a) and (b) from each main question.

(iii) Total marks for each sub-question is 10 marks.

1. (a) The Dehradun ISBT consists of 30 platforms and 100 buses. The platforms are numbered from 1 to 30, and buses have numbers from 1 to 100. At ISBT, authorities mentioned that platform numbers divisible by 4 and 6 are platforms for Delhi, and the prime numbers of the bus from 20 to 50 are commuting from Dehradun to Delhi. The platform numbers divisible by 5 and 15 are the platforms for Chandigarh, and the prime numbers of the bus from 60 to 90 are commuting from Dehradun to Chandigarh. Develop a Python program to help Aryan identify Delhi platforms and buses going from Dehradun to Delhi. Develop a Python program to help Ryan identify Chandigarh platforms and buses going from Dehradun to Chandigarh.

(CO3)

P. T. O.

OR

(b) The printing presses are charging less for bulk paper printing. The charges are as follows :

- (i) Pages from 1 to 100 are ₹ 4 per page.
- (ii) Pages from 101 to 200 are ₹ 3 per page.
- (iii) Pages from 201 to 300 are ₹ 2 per page.
- (iv) More than 300 are at ₹ 1.5 per page.

Ram and Shyam are regular customers of the printing press. The bulk page printing order that Ram gave a few days ago is [225, 348, 765, 178, 278, 476, 78], and Shyam gave one that is [456, 234, 167, 98, 365]. Develop a Python program to calculate and print the total amount for printing orders given by Ram and Shyam. (CO3)

2. (a) Explain the similarities and differences between list and tuple with examples. (CO1)

OR

(b) Discuss the Python data types and arithmetic operators with examples.

(CO1)

3. (a) The smart Apple farm consists of a weather sensor to monitor the temperature of the farm and a soil sensor to measure the current pH of the soil. Soil pH is a measure of the neutrality, acidity, or alkalinity of the soil. Soils can be classified according to their pH value :
- (i) 6.5 to 7.5 is neutral
 - (ii) More than 7.5 is alkaline
 - (iii) less than 6.5 is acidic

The data generated from sensors is stored in Python list. The last seven days temperature data generated by the weather sensor in Fahrenheit is [65, 112, 76, 54, 59, 92, 82], and the soil pH data generated from the soil sensor is [8.2, 6.3, 7.6, 5.9, 8.2, 4.9, 6.8]. Develop a menu driven Python program to help farmers convert weather data from Fahrenheit to Celsius and classify the alkaline or acidic pH value of the soil into a separate list. (CO6)

OR

- (b) For the University's cultural and sports events, the students are registered through an online portal, and their values are stored on a Python set data structure. The following set of students are registered for cultural {Prateek, Yash, Samera, Impana, Anurag, Kushi} and sports {Jainkar, Yash, Samera, Smaya, Ayush, Kushi}. Develop a menu driven Python program to perform the following operations and write the output after each operation :
- (i) To add the name of the new student to any or both of the events.
 - (ii) To display the names of the students who are registered for cultural or sports events.
 - (iii) To display the names of the students who are registered for both cultural and sports events.
 - (iv) To display the names of the students who are registered only for cultural events.
 - (v) To display the names of the students who are registered for either cultural or sports events but not in both. (CO6)

P. T. O.

(4)

4. (a) Malcolm uses SpeechTexter to convert his spoken words into written text in real-time. Once, he said, "Education is the passport to the future comma for tomorrow belongs to those who prepare for it today." The sentence is stored in a variable named `str`. Develop a Python program to perform the following operations and write the output after each operation :

- (i) To replace a word 'comma' by ','.
- (ii) To make first character of each word uppercase and count total number of uppercase letters.
- (iii) To count the number of characters (including special symbols).
- (iv) To count the number of vowels.
- (v) To count non-alphabet letters.

(CO5)

OR

- (b) Shashank, an NLP engineer, once web scraped the following sentence from the ISRO website "Chandrayaan 3, which is India's most awaited lunar expedition, touched the moon's surface on August 23, 2023, at 6 : 04 p.m. IST with a speed of 1.68 km/s." The sentence is stored in a variable named `text`. Develop a Python program to perform the following operations and write the output after each operation :

- (i) To count total number of numeric in the given text.
- (ii) To count the number of white spaces.
- (iii) To search the text "IST", if it is found, return its position.
- (iv) To count the total number of non-alphanumeric characters in the given text.
- (v) To split the text on the separator ',' and store in a separate list.

(CO5)

5. (a) The Metropolitan Cities of India are stored on a Python tuple data structure. The names of the cities are stored in a variable named city ('Mumbai', 'Delhi', 'Kolkata', 'Chennai', 'Bangalore', 'Hyderabad'). Develop a menu driven Python program for the following operations on Python tuple data structure :

- (i) To add a new city name to a Tuple.
- (ii) To update a city name on a Tuple.
- (iii) To remove a city name from a Tuple.
- (iv) To display all city names.

(CO3)

OR

(b) The country's name and its capital city name are stored in a Python dictionary data structure. The country_capitals = {
"India", "New Delhi", "United States" : "Washington D.C.", "Russia" :
"Moscow", "England" : "London"
}

Develop a menu driven Python program for the following operations on Python dictionary data structure :

- (i) To add a new element to a dictionary.
- (ii) To Update an element in a dictionary.
- (iii) To remove an element from a dictionary.
- (iv) To display dictionary elements.

(CO4)

Roll No.

TCS-342

**B. TECH. (AI&DS) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

INTRODUCTION TO STATISTICAL DATA SCIENCE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Give working definition of Data Science. Define the term Big data.

(CO1)

OR

- (b) Explain at least *four* real-world applications driven by Data Science.

(CO2)

2. (a) Explain Probability – $P(x)$ as generic Lebesgue measurable rather than relative frequency. Determine if are valid Distribution Functions (D. F.).

(CO3)

OR

- (b) Denote closed form expressions and illustrate (graphically) PMF/PDF of random variables : Bernoulli, Poisson, Exponential, Uniform. (CO2)

P. T. O.

(2)

3. (a) Given random trials in tossing fair coin; obtain, if. Derive Binomial random variable from Bernoulli random variable and give its mean and variance. (CO1)

OR

- (b) Plot p. m. f. and D. F. of denoting outcome in rolling fair dice, evaluate the mean and variance for this random variable. Given indicator random variable, derive D. F. (CO2)
4. (a) Given is Gaussian distributed with mean and variance of, find mean and variance of, plot p.d.f. and D.F. of. What is the support set of X and of Y? (CO2)
- (b) Give physical reasons and explain which random variable best describes the following data. Also depict p.d.f./p.m.f. in each answer :
- (i) arrival of people at bus-stop,
 - (ii) ratings from online reviews of mediocre liked product/movie,
 - (iii) battery life of cell phone,
 - (iv) c.g.p. a distribution in class of 20 students,
 - (v) predicting stock value with no prior knowledge, other than *max* and *min* values (CO3)
5. (a) Obtain correlation co-efficient for ten data-points sampled from time-series : where in this case we choose, 1. Formulate the general correlation co-efficient and what trend does it indicate? (CO3)

OR

- (b) Given primary data X, obtain population mean, sample median, sample mean, sample variance, population variance. Also show. (CO2)

Roll No.

TCS-343

B. TECH. (CSE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

MATHEMATICAL FOUNDATIONS FOR ARTIFICIAL INTELLIGENCE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) (i) Show that there is a unique linear map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ for which

$$T(3, 1) = (2, -4) \text{ and } T(1, 1) = (0, 2). \quad (\text{CO1})$$

(ii) Find a formula for T , i. e., find $T(a, b)$

(iii) Find $T^{-1}(5, -3)$.

OR

(b) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear mapping defined by

$$T(x, y, z) = (x + 2y - z, y + z, x + y - 2z).$$

Find a basis and the dimension of the image U of T . (CO1)

P. T. O.

2. (a) Consider the following subspaces of $\mathbb{R}^4$: (CO1)

$U = \{(a, b, c, d) : b + c + d = 0\}$ and $W = \{(a, b, c, d) : a + b = 0, c = 2d\}$. Then, find the dimension and a basis of U and W .

OR

- (b) Determine whether the polynomials u, v, w are dependent where :

(CO1)

$$u = t^3 + 4t^2 - 2t + 3,$$

$$v = t^3 + 6t^2 - t + 4,$$

$$w = 3t^3 + 8t^2 - 8t + 7.$$

3. (a) Show that $u_1 = (1, 2, 5), u_2 = (1, 3, 7),$ and $u_3 = (1, -1, -1)$ do not span $\mathbb{R}^3$.

(CO1)

OR

- (b) Find a symmetric orthogonal matrix P whose first row is $\left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$.

(CO2)

4. (a) Consider the basis $B = \{(1, 1, 0), (1, 2, 3), (1, 3, 5)\}$ of $\mathbb{R}^3$. Find the matrix A which represents the usual inner product on $\mathbb{R}^3$ with respect to the basis B . (CO2)

OR

- (b) Let V be the vector space of polynomials $f(t)$ of degree ≤ 2 with inner

product defined by $\langle f, g \rangle = \int_{-1}^1 f(t)g(t)dt$. Find $\langle f, g \rangle$ where

$$f(t) = t + 2 \text{ and } g(t) = t^2 - 3t + 4. \quad (\text{CO2})$$

(3)

5. (a) Let V be a vector space of real continuous functions on the interval $a \leq t \leq b$. Show that the following is an inner product on V . (CO2)

$$\langle f, g \rangle = \int_a^b f(t)g(t)dt$$

OR

- (b) Find an orthonormal basis of the subspace W of $\mathbb{R}^5$ spanned by : (CO2)

$$v_1 = (1, 1, 1, 0, 1),$$

$$v_2 = (1, 0, 0, -1, 1),$$

$$v_3 = (3, 1, 1, -2, 3),$$

$$v_4 = (0, 2, 1, 1, -1).$$

Roll No.

TCS-351

B. TECH. (CSE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
FUNDAMENTAL OF CLOUD COMPUTING AND BIG DATA

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Healthcare systems around the world are unsustainable. The global pandemic has only exacerbated existing staffing and physician shortages, increased cost pressures, and increased service backlog and capacity issues. For global virtual care company, Teladoc Health, the partnership between HITRUST and Microsoft enables data security, ensures regulatory compliance, and helps their platform maintain and grow trust with partners and patients. Evaluate Cloud Computing Vision from perspective of above mentioned organization. (CO1)

OR

- (b) When Provincial Health Services Authority (PHSA) began seeing the need to deliver faster, more comprehensive health insights to healthcare workers across British Columbia, it took action. Old, unstandardized requests were slowing data analysis and the sharing of insights. To

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standardize and speed data access, PHSA moved its healthcare analytics platform to Microsoft Azure, creating a readily accessible data analysis platform that is scalable and secure. Data requests that once took many months now take only days. Because of the sensitive nature of health data, PHSA has worked closely with Microsoft to make sure health data never leaves Canada and is kept highly secure. Evaluate Characteristics of Cloud Computing from perspective of above mentioned organization. (CO2)

2. (a) Hamad Medical Corporation (HMC) is the leading healthcare provider in Qatar. It was looking for a security information and event management (SIEM) solution. "Our vision was to deliver security without compromising access and usability. We also wanted to ensure patient and corporate confidentiality at all times," says Richard Storey, Executive Director / Acting Chief of Health Information and Communication Technology at HMC. With Microsoft Sentinel, HMC gained 360-degree, real-time visibility over cyber security, making proactive responses faster for supporting staff to access information. Analyze Cloud computing vs. Cluster computing vs. Grid computing from perspective of above mentioned organization. (CO2)

OR

- (b) Growing healthcare organization MultiCare Health System teamed up with Microsoft and Microsoft partner 3Cloud to meet changing demands in health data access prompted by the use of fitness trackers and phones. MultiCare wanted a scalable, interoperable data exchange framework to connect providers, patients, and payers-key for success in

population health and value-based care. It also required an adaptable solution. Working with Microsoft FHIR experts and partnering with 3Cloud, MultiCare built a unique, fully managed solution using Microsoft Azure. The results have surpassed expectations : in three months, MultiCare has seen a significant reduction in administrative burden with a \$50 per-chart cost savings and a 65 percent reduction in labor. Analyze the Importance of Open Standards for digital age technologies from the perspective of above mentioned organization.

(CO3)

3. (a) Mercy, one of the largest healthcare systems in the United States, aided its digital transformation by using Microsoft Azure to decrease operational costs, improve the patient experience, and give its providers better access to data. The company wanted to modernize its infrastructure and extract more value from its sizeable data archives to keep up with evolving patient and doctor expectations. With support from Microsoft, Mercy identified business outcomes and undertook a digital transformation to achieve them. On Azure, the company improved efficiency, connected previously siloed data, reduced patient stay durations, and built a foundation for ongoing innovation. Analyze IAAS from the perspective of above mentioned organization.

(CO2)

OR

- (b) When Centro Hospitalar de Trás-os-Montes e Alto Douro needed a way to protect sensitive, highly confidential patient data and applications in its hybrid environment, it chose Microsoft Security solutions. The

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challenges in scaling its on-premises data warehouse. Additionally, the health network wanted to explore AI, predictive analytics, and other innovative technologies while maintaining a high level of security for patient data. It deployed Azure Synapse Analytics to scale up its data integration and storage capabilities in a secure cloud system. By creating this system, Health First lowered costs, improved workload processing times, and offered better data access for providers. Analyze Cloud Computing comparison with traditional computing architecture from the perspective of above mentioned organization. (CO3)

Roll No.

TCS-392

B. TECH. (CSE AND SPECIALIZATIONS)

(THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

INTRODUCTION TO CRYPTOGRAPHY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What do you understand by the term Cryptography and what are its applications ? Elaborate. Further, provide the block diagram depicting essential elements of conventional encryption and explain the concepts like Key, Plaintext and Ciphertext using equations. (CO1)

OR

- (b) How do you differentiate between Cryptanalysis and Brute-force attack ? What are the advantage and limitations of each approach ? Further, elaborate with the help of a table the Cryptanalysis based on types of attack on what is known to cryptanalyst. (CO1)
2. (a) List out and define any five security services that can be provided to the users to securely transmit the message from source to destination. What do you understand by Polyalphabetic Ciphers ? Explain the concept with a typical example of Vigenere Cipher. (CO1)

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OR

- (b) (i) Briefly explain the concept of Caesar Cipher. Encrypt the following message using the default shift key of Caesar cipher : (CO1)

beware of the jaws that bite the claws that catch

- (ii) You intercept the following ciphertext :

GTRG UQFSYJI FY YMJ MNLMBFD SZRGJW TSJ

You also analyze that this ciphertext is generated using a Caesar Cipher and the key used is between 3 to 6. Can you identify exact key and find out meaningful plain text ?

3. (a) Consider two bits of plaintext (PT). Based on the Fiestel Block Cipher explain with the help of an example the reversible and irreversible mapping of PT to CT. (CO2)

Further for two-bit PT, answer the following :

- (i) How many plaintext blocks are possible ?
- (ii) How many different transformations are possible ?
- (iii) What is the key length in bits desirable to thwart statistical attack ?
- (iv) What do you conclude from the required key-length ?

OR

- (b) (i) Elaborate on the concept and types of transposition techniques in Cryptography.
- (ii) Consider a message "eagle docking pearl harbour december the fourth". Make use of rail fence technique with three rails to convert it to Ciphertext. (CO2)

(3)

4. (a) What is the difference between an unconditionally secure cipher and a computationally secured cipher ? Explain the algorithm that is considered unconditionally secure in brief and emphasize its limitation for practical usage. (CO2)

OR

- (b) Consider a workstation connected to Internet through LAN in a typical business environment, list potential locations for confidentiality attacks. Further, with the help of a neat figure explain the concept of Link level and End-to-End encryption. (CO2)
5. (a) (i) Provide the structure of Key generation in S-DES using block diagram and explain each of the blocks. (CO2)
- (ii) Given the Key (in Binary) 10001 11001 and following P-boxes. Provide step-wise derivation of sub-keys K1 and K2 :

P10									
3	5	2	7	4	10	1	9	8	6

P8							
6	3	7	4	8	5	10	9

OR

- (b) Elaborate the concept of avalanche effect in DES with one or two typical values justifying your explanation. Further, elaborate on two important factors that are used to assess the strength of DES algorithm.

(CO2)